

# DHEERAJ SONGARA

Graduate Student | Int. Ph.D. in Neuroscience  
National Centre for Biological Sciences, Bengaluru

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**RESEARCH INTEREST** *To explore the neural basis of memory formation and engram theory through an integrative approach combining electrophysiology, in vivo imaging, transcriptomics, and behavioral neuroscience.*

## WORK EXPERIENCE

**Research Scholar** · *National Centre for Biological Sciences, Bengaluru* *Sep 2021 – Present*

- ▶ Investigating pyramidal neuron subtypes in the adult mouse hippocampus.
- ▶ Combining ex vivo electrophysiology, single-cell RNA sequencing, and behavioural assays to study neuronal subtypes.
- ▶ Using chemogenetics and viral injections for targeted neuronal manipulation.

**Research Scholar** · *National Centre for Biological Sciences, Bengaluru* *Dec 2018 – Sep 2021*

- ▶ Studied amygdalar dysfunction in Fragile X Syndrome and the role of astrocytic glutamate uptake in stress.
- ▶ Conducted patch-clamp recordings, auditory fear conditioning, and chronic stress paradigms in rodent models.

**Industry Intern** · *CIPLA, Pithampur, India* *Jul 2014 – Dec 2014*

## EDUCATION

**Int. Ph.D. — Neuroscience** · *National Centre for Biological Sciences, Bengaluru* *Jul 2018 – Present*

**B.Sc. — Biotechnology** · *Choithram College of Professional Studies, Indore* *Jul 2015 – 2018*

## TECHNICAL SKILLS

### Electrophysiology & Imaging

- ▶ Whole-cell patch clamp (brain slices & cultured neurons)
- ▶ Two-photon calcium imaging analysis (Suite2p)
- ▶ In vitro calcium imaging
- ▶ Confocal & fluorescence microscopy
- ▶ Morphological analysis (Sholl's analysis, Neurolucida)

### Molecular & Cellular Biology

- ▶ Immunofluorescence, western blotting, qPCR
- ▶ FACS & MACS for cell sorting
- ▶ Neuronal & HEK cell culture, viral production, transfection

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| <b>Animal Handling &amp; Behavioural Neuroscience</b> <ul style="list-style-type: none"> <li>▶ Mouse &amp; rat handling, breeding, genotyping (PCR, gel electrophoresis)</li> <li>▶ Stereotaxic viral injections, perfusion, brain slicing</li> <li>▶ Behavioural assays: OFT, EPM, Y-maze, NOR, NLR, fear conditioning</li> </ul> | <b>Genomics &amp; Computational Analysis</b> <ul style="list-style-type: none"> <li>▶ Single-cell RNA sequencing, bulk RNA sequencing</li> <li>▶ Python — electrophysiology &amp; single-cell analysis pipelines</li> <li>▶ Software: ImageJ, Imaris, FlowJo, LasX Confocal, AnyMaze, FACSDiva</li> </ul> |
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## PUBLICATIONS

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1. **Songara D. & Ghosh H.** *Manipulation of CA1 neuronal subtypes through Cre-mediated viral delivery in mice.* [BioRxiv 05.08.723440](https://doi.org/10.1101/05.08.723440)
2. **Songara D. & Ghosh H.** *Memory encoding and retrieval are executed by molecularly specialized subsets within the engram ensembles. (under submission)*
3. **Fernandes et al., 2021.** *Correction of Amygdalar Dysfunction in a Rat Model of Fragile X Syndrome.* [Cell Reports, 37, 109805.](https://doi.org/10.1038/s41467-021-10980-5)

## TEACHING & PRESENTATIONS

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- ▶ Poster presentation scheduled at FENS Forum 2026, Spain.
- ▶ Presented at NGN24 (No Garland Neuroscience), IISER TVM.
- ▶ Teaching Assistant – Neuroscience Course, NCBS (2 years).